

# Static Gesture Classification — Run Report

Generated 2026-05-13 18:47:34

## Data paths

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| Training data path 1  | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials       |
| Training data path 2  | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials       |
| Training data path 3  | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials     |
| Training data path 4  | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials     |
| Training data path 5  | /home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials    |
| Training data path 6  | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials      |
| Training data path 7  | /home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 35 trials      |
| Training data path 8  | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials     |
| Training data path 9  | /home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials    |
| Training data path 10 | /home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials      |
| Training data path 11 | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials     |
| Training data path 12 | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials    |
| Training data path 13 | /home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials       |
| Training data path 14 | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials |
| Training data path 15 | /home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials     |
| Training data path 16 | /home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 35 trials    |
| Total training paths  | 16 · 559 trials total                                                  |
| Report output dir     | /home/jestin/ThesisRepo/ML/NewReports/StaticReports                    |

## Sensor selection

|                |                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| Hands          | left, right                                                                                                         |
| Segments       | palm_mid, thumb, index, middle, ring, pinky, wrist                                                                  |
| Modalities     | YPR, Accel, Flex                                                                                                    |
| Sensor columns | 176 channels                                                                                                        |
| Classes        | 7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman |

## Preprocessing, features & split

|                   |                                                                 |
|-------------------|-----------------------------------------------------------------|
| Resample steps    | 90                                                              |
| Butterworth       | on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz                      |
| Normalisation     | minmax                                                          |
| Feature mode      | stats+fft → feature dim 9504                                    |
| Split / seed / CV | random_state=42 · CV: LOUO (16 folds, every user held out once) |
| Sample counts     | train (post-aug)=559 · LOUO-evaluated samples=559               |

## Augmentation (training only)

|                        |                        |
|------------------------|------------------------|
| Enabled                | no                     |
| Copies per train trial | 4                      |
| Jitter                 | off · sigma=0.01       |
| Amplitude scale        | on · range=(0.9, 1.1)  |
| Time warp              | off · range=(0.9, 1.1) |

## Classifiers — LOUO cross-validation

| Algorithm     | CV mean (LOUO, 16 folds) | CV std | CV min | CV max |
|---------------|--------------------------|--------|--------|--------|
| Random Forest | 0.9339                   | 0.0800 | 0.7714 | 1.0000 |

## Per-class CV F1 (across all LOUO folds)

| Class              | Random Forest | Support |
|--------------------|---------------|---------|
| Double_Closed_Fist | 0.931         | 83      |
| Double_Nothing     | 0.809         | 75      |
| Double_Okay        | 0.981         | 80      |
| Double_Open_Palm   | 0.982         | 80      |
| Double_Phone       | 0.915         | 80      |
| Double_Pistol      | 0.910         | 81      |
| Double_Spiderman   | 0.994         | 80      |
| macro avg          | 0.932         | 559     |

## Per-user accuracy (each user as held-out fold)

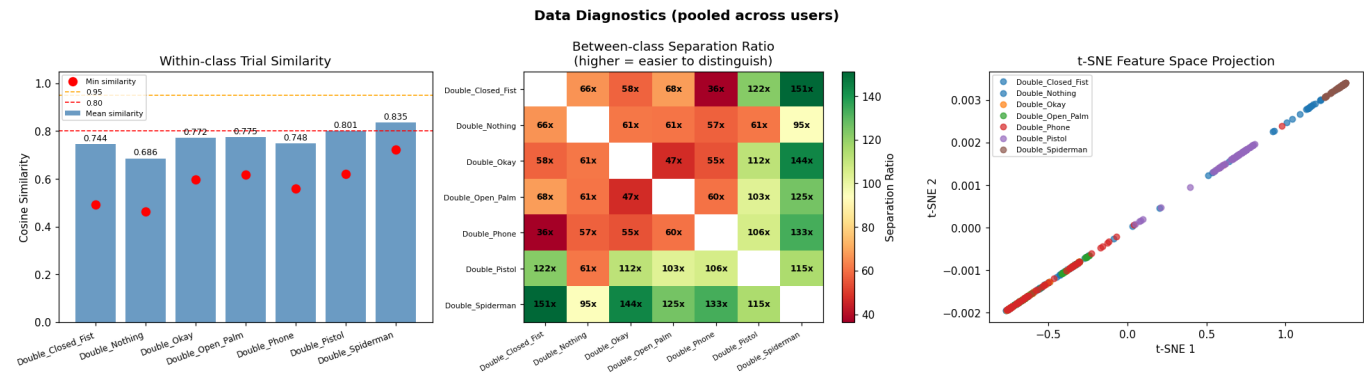
| User     | n trials | Random Forest |
|----------|----------|---------------|
| 1_Alan   | 35       | 1.000         |
| 2_Alex   | 35       | 0.800         |
| 3_Anghad | 35       | 0.943         |
| 4_Daniel | 35       | 0.943         |

|                     |    |       |
|---------------------|----|-------|
| 5_Georgia           | 35 | 1.000 |
| 6_Harry             | 35 | 0.829 |
| 7_Henry             | 35 | 0.771 |
| 8_Jestin            | 36 | 1.000 |
| 9_Joselyn           | 30 | 1.000 |
| 10_Josh             | 35 | 0.829 |
| 11_Mansh            | 36 | 1.000 |
| 12_Marcus           | 35 | 1.000 |
| 13_Nat              | 37 | 1.000 |
| 14_StephenV2        | 35 | 0.971 |
| 16_Bella            | 35 | 0.886 |
| 17_Raquel           | 35 | 0.971 |
| Mean (across users) |    | 0.934 |
| Std (across users)  |    | 0.080 |

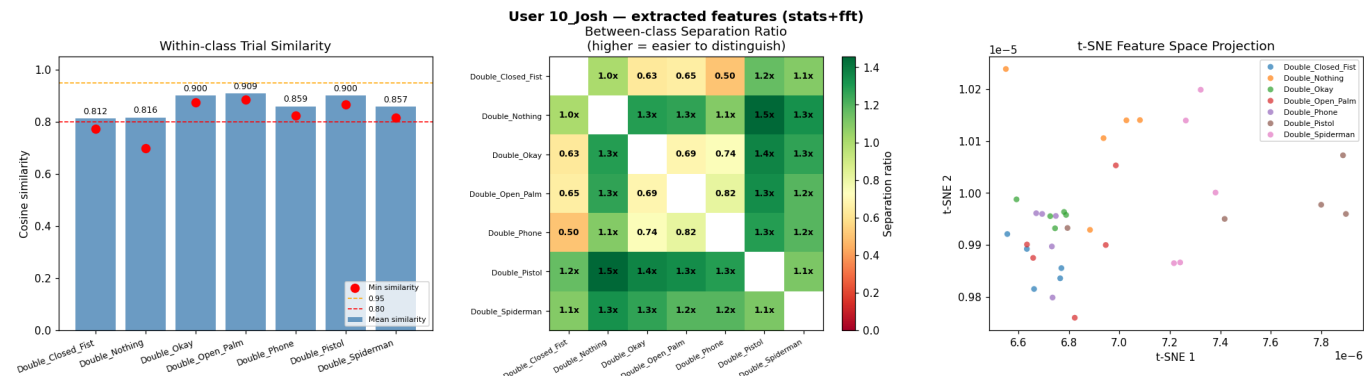
Appendix — Data Diagnostics

17 figures: pooled across users + one per detected user (extracted-feature views).

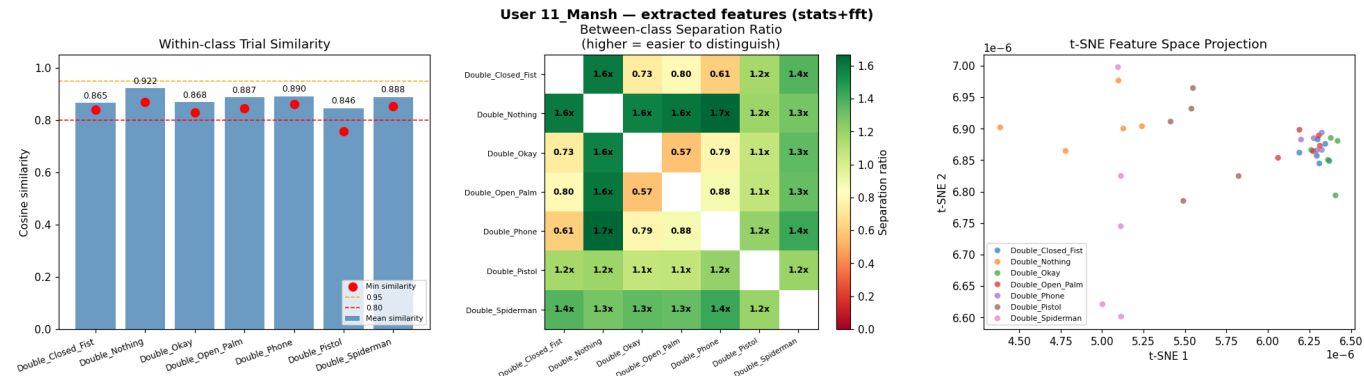
Pooled diagnostics — extracted features (across all users)



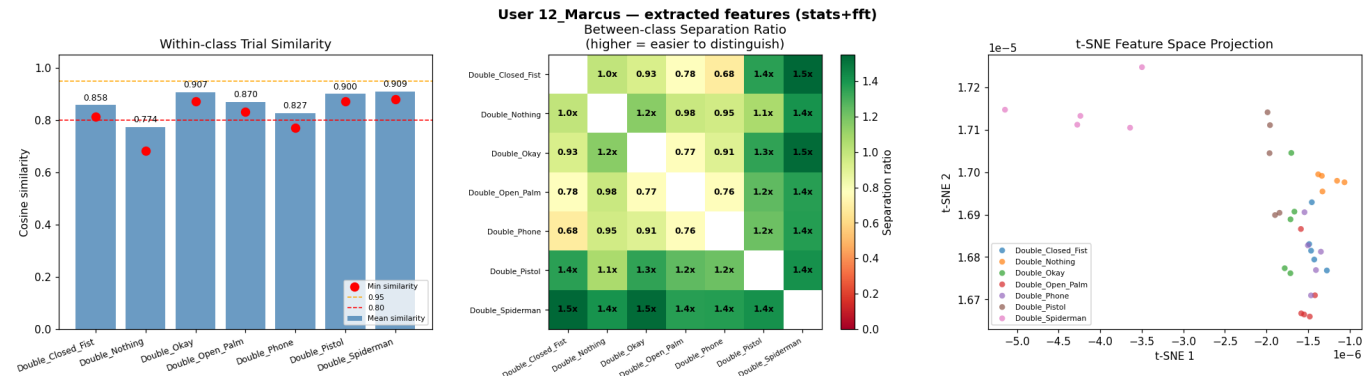
User 10\_Josh — extracted features (stats+fft)



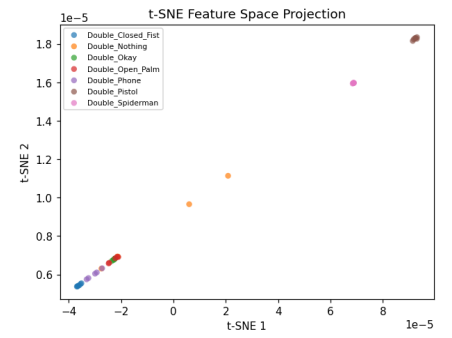
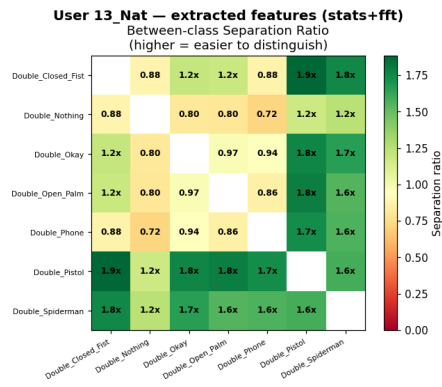
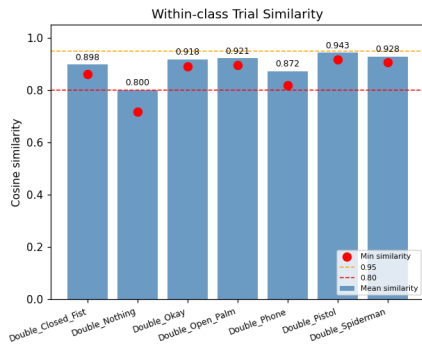
User 11\_Mansh — extracted features (stats+fft)



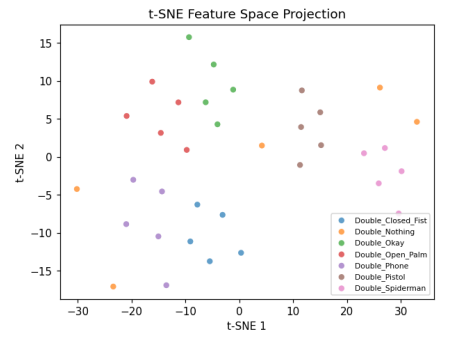
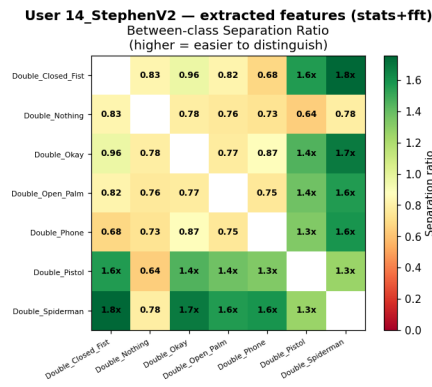
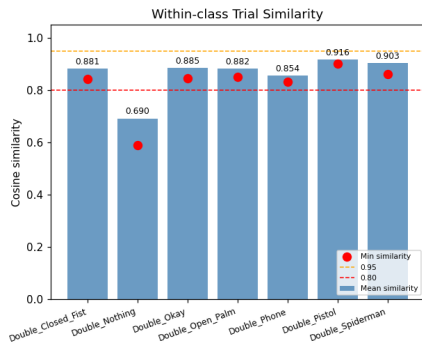
User 12\_Marcus — extracted features (stats+fft)



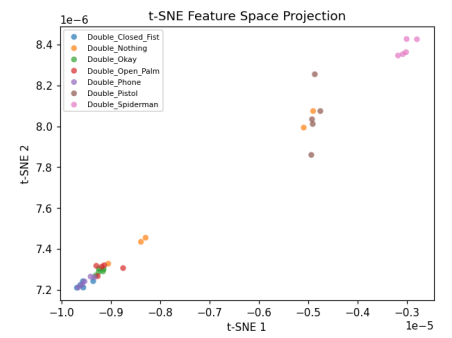
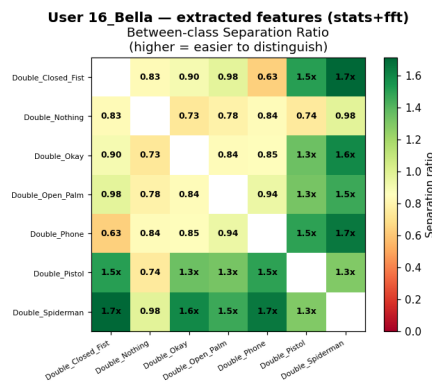
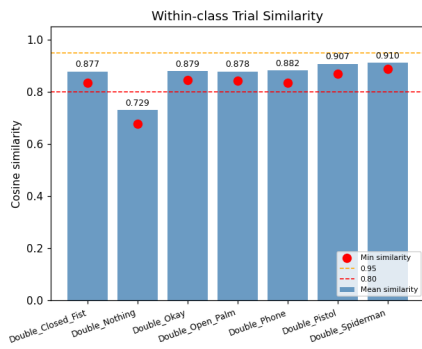
User 13\_Nat — extracted features (stats+fft)



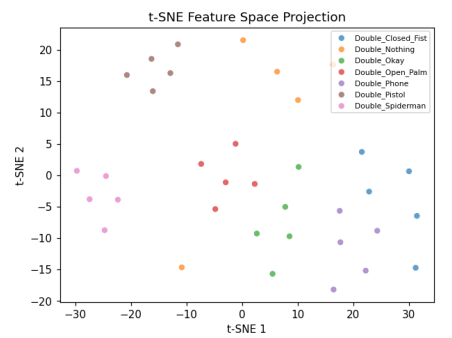
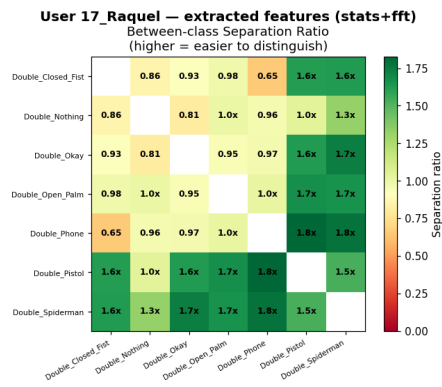
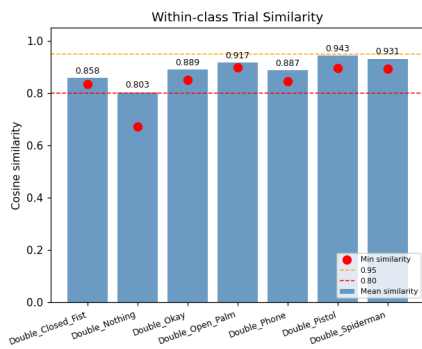
## User 14\_StephenV2 — extracted features (stats+fft)



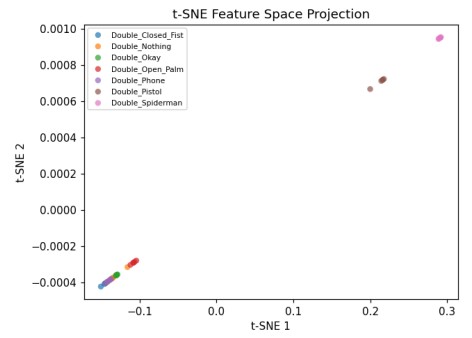
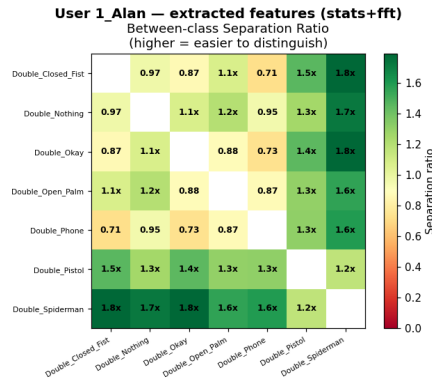
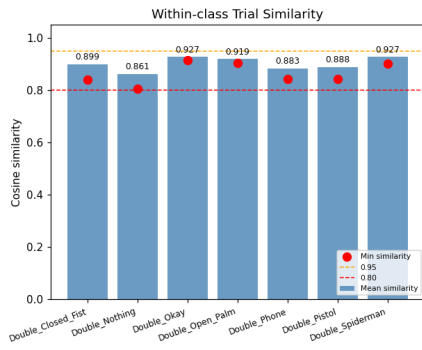
## User 16\_Bella — extracted features (stats+fft)



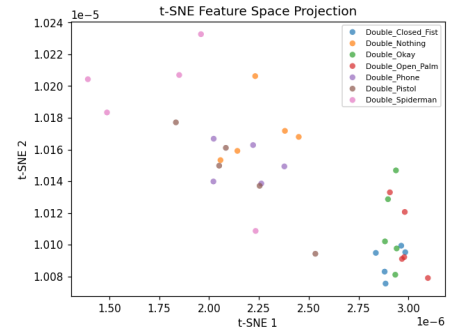
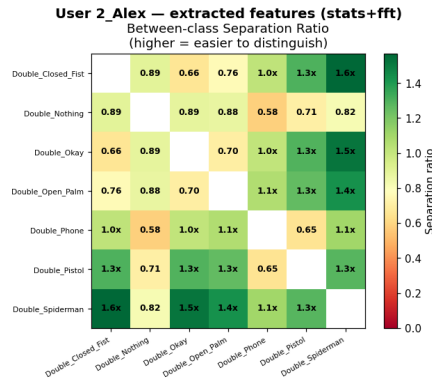
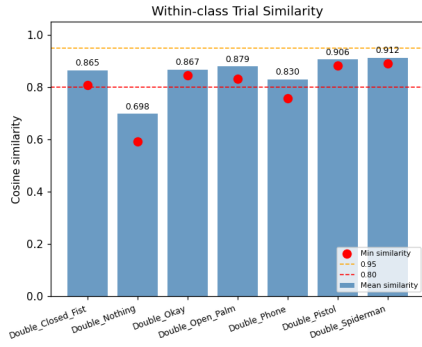
## User 17\_Raquel — extracted features (stats+fft)



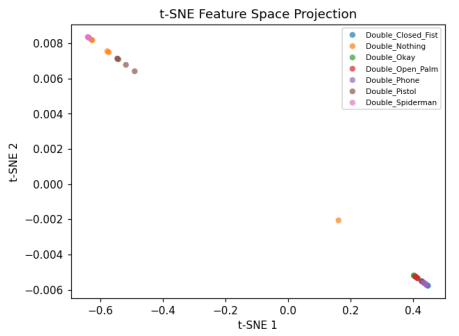
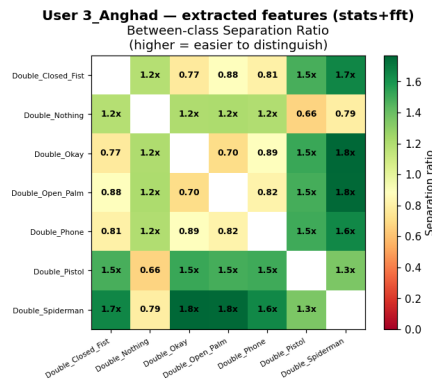
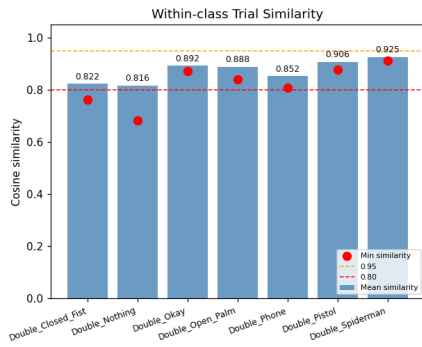
## User 1\_Alán — extracted features (stats+fft)



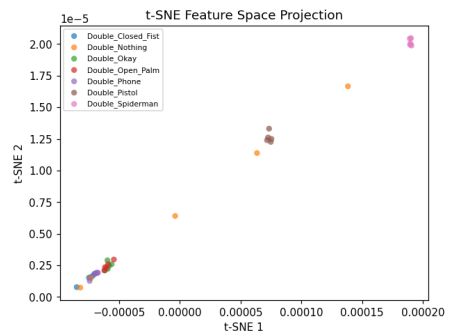
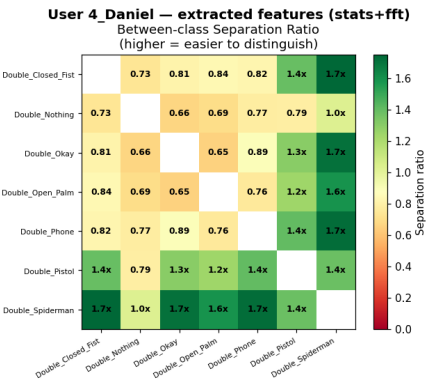
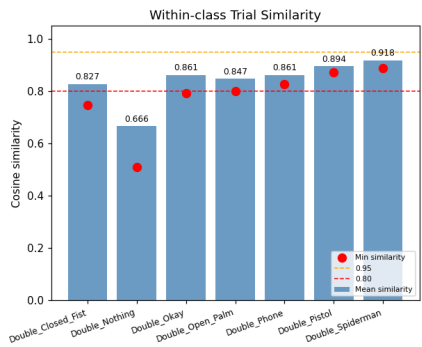
## User 2\_Alex — extracted features (stats+fft)



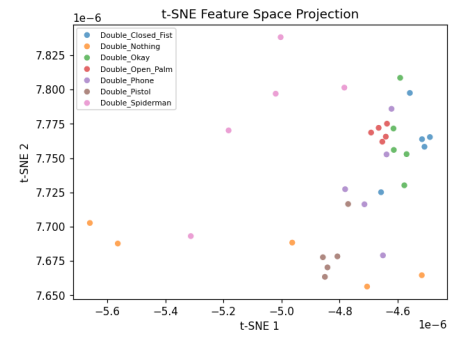
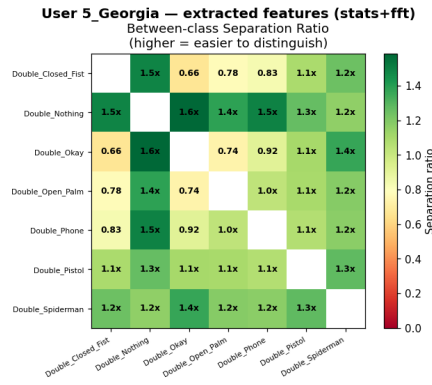
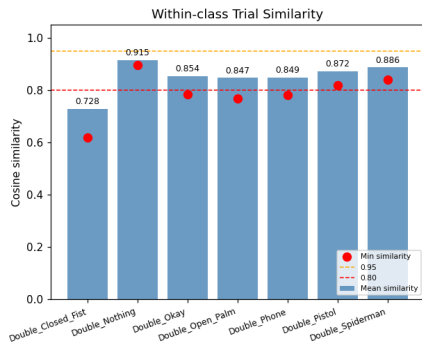
## User 3\_Anghad — extracted features (stats+fft)



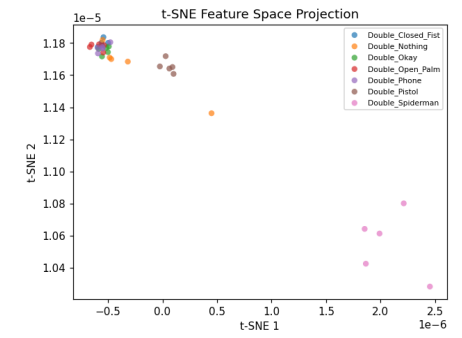
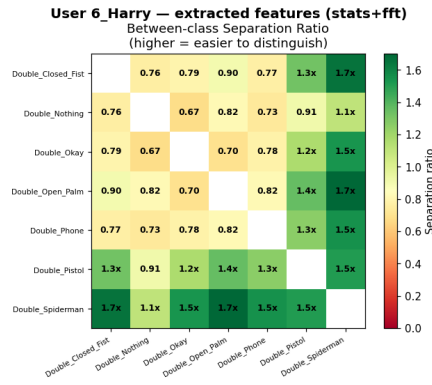
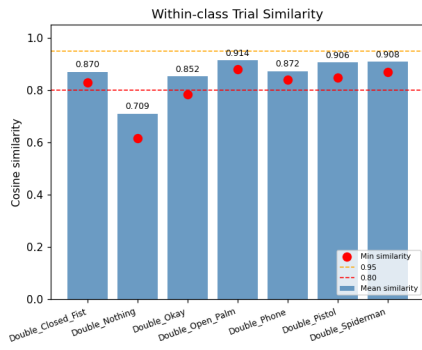
## User 4\_Daniel — extracted features (stats+fft)



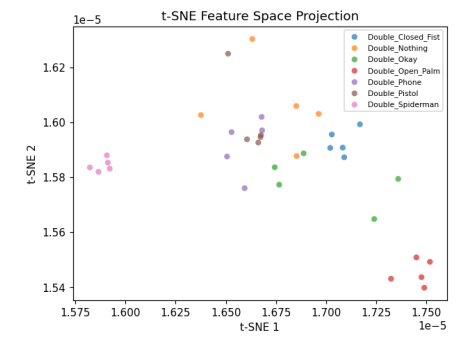
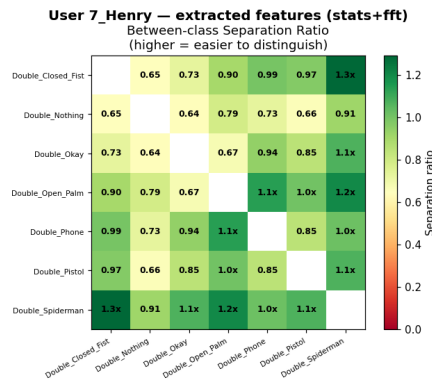
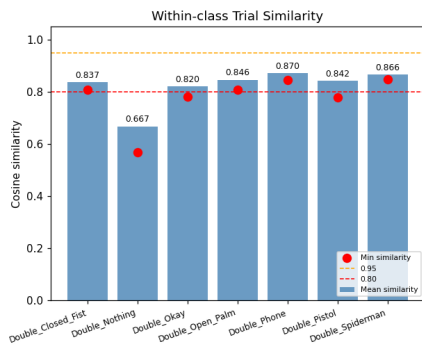
## User 5\_Georgia — extracted features (stats+fft)



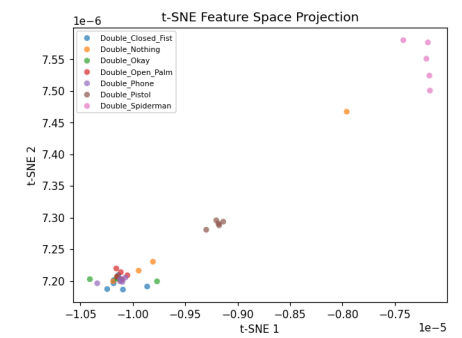
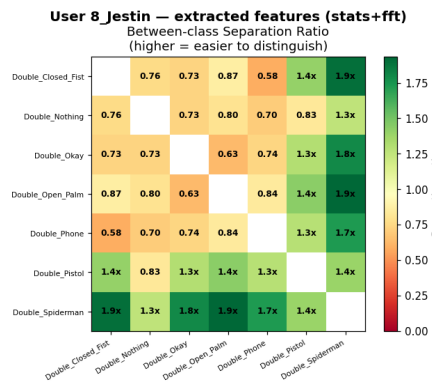
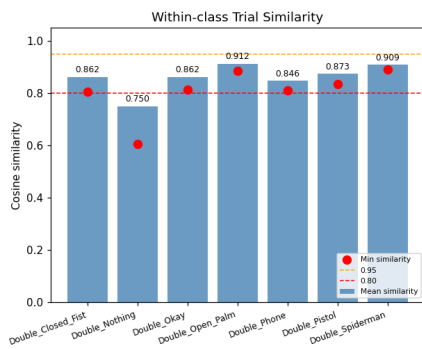
## User 6\_Harry — extracted features (stats+fft)



## User 7\_Henry — extracted features (stats+fft)



## User 8\_Jestin — extracted features (stats+fft)



## User 9\_Joselyn — extracted features (stats+fft)

